MIRROR: A Hierarchical Benchmark for Metacognitive Calibration in Large Language Models

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Abstract

We introduce MIRROR (Metacognitive Integrity and Recursive Reasoning in Operationalized Response), a benchmark comprising eight experiments across four metacognitive levels that evaluates whether large language models can use self-knowledge to make better decisions. We evaluate 16 models from 8 labs across approximately 250,000 evaluation instances using five independent behavioral measurement channels. We find two phenomena with direct implications for agentic deployment: (1) compositional self-prediction fails universally—the Metacognitive Convergence Index is exactly 0.000 across all 15 models tested, and (2) models possess accurate domain-specific self-knowledge but systematically fail to translate it into appropriate agentic action-selection—external metacognitive control reduces the Confident Failure Rate from 0.562 to 0.214, a 62% reduction. Providing models with their own calibration scores produces no significant improvement ($p > 0.05$); only architectural constraint is effective. The Knowing-Doing Index is negative for 9 of 10 models in the capstone experiment. These findings challenge the assumption underlying current agentic architectures that accurate self-knowledge enables self-correction, and suggest that external metacognitive scaffolding—not improved self-knowledge—is the path to safer autonomous AI systems. MIRROR and all data are publicly available.

1 Introduction

The deployment of large language models (LLMs) as autonomous agents presupposes a form of metacognitive competence: the model must not only know *what* it knows, but must translate that awareness into appropriate action—proceeding when confident, escalating to tools or human review when uncertain. This assumption underlies agentic architectures built on self-reflection loops, confidence-based routing, and Constitutional AI self-monitoring [Perez et al., 2022]. It is foundational to the enterprise of agentic AI safety. It has never been systematically tested.

Prior work has studied LLM calibration [Kadavath et al., 2022, Kapoor et al., 2024], uncertainty estimation [Lin et al., 2024], and situational awareness [Laine et al., 2024]. But these address different questions: calibration measures whether confidence matches accuracy; situational awareness measures whether models know *what they are*. Nobody has systematically tested whether models can *use* their self-knowledge to make better decisions—whether the bridge from metacognitive *monitoring* to metacognitive *control* [Nelson and Narens, 1990] is intact.

We introduce MIRROR, a hierarchical benchmark spanning four metacognitive levels (atomic self-knowledge, cross-domain transfer, compositional prediction, and adaptive self-regulation), eight experiments, and five independent behavioral measurement channels. We evaluate 16 models from 8 labs across approximately 250,000 evaluation instances.[1]

Three findings emerge. First, compositional self-prediction fails universally: the Metacognitive Convergence Index (MCI) is exactly 0.000 across all 15 models tested—models that calibrate accurately in individual domains completely fail to compose those calibrations for multi-domain

[1]Computed as: Exp1 (5,000 $\times$ 5 channels $\times$ 16 models) + Exp4 (320 $\times$ 2 conditions $\times$ 15 models) + Exp9 (597 $\times$ 4 conditions $\times$ 3 paradigms $\times$ 10 models) + Experiments 2, 3, 5, 6, 8 ($\sim$30,000 total). See Table 3 for per-experiment counts.

Table 1: Comparison of MIRROR with prior benchmarks for LLM self-knowledge and metacognition.

Feature	SAD (Laine 2024)	Barkan et al. (2025)	Kadavath et al. (2022)	Steyvers et al. (2025)	MIRROR (Ours)
Venue	NeurIPS D&B '24	NeurIPS WS '25	arXiv	arXiv	NeurIPS D&B '26
Focus	Situational awareness	Self-capability prediction	Calibration (P(IK))	Metacognition + uncertainty	Metacognitive cal. → agentic action
Models	16	∼15	∼10	2 families	16
Task categories	7	3	—	6 datasets	8 experiments
Total items	13K questions	∼1K tasks	—	12K+ items	∼250K records
Metacognitive levels	1 (awareness)	1 (confidence)	1 (calibration)	1 (calibration)	4 (L0–L3)
Behavioral channels	1 (MC/free)	1 (verbal)	1 (verbal)	1 (verbal)	5 (wager, opt-out, tool, diff., natural)
Agentic connection	No	Yes (SWE-bench)	No	No	Yes (4 cond., 3 par.)
Headline finding	Partial sit. awareness	LLMs over-confident	*Mostly* calibrated	Task-specific metacognition	Knowing-doing gap (62% CFR reduction)
Public dataset	Yes	No	No	No	Yes

tasks. Second, the knowing-doing gap: models possess accurate self-knowledge but do not act on it. External constraint reduces the Confident Failure Rate (CFR) from 0.562 to 0.214 (62% reduction); providing models with their own calibration scores produces no significant improvement. Third, self-knowledge is domain-atomic: the Transfer Influence Index ranges from 0.019 to 0.175 across 11 models, indicating that calibration does not meaningfully cross domain boundaries.

Contributions.

- The MIRROR benchmark: 8 experiments across 4 metacognitive levels, 5 behavioral channels, 5,000 domain-stratified questions, and 597 composite agentic tasks, publicly released with all protocols and analysis code.
- Two empirical findings—universal compositional failure (MCI = 0.000) and the knowing-doing gap (62% CFR reduction under external control)—that challenge assumptions about LLM self-knowledge.
- Quantification of the knowing-doing gap's cost: in a 1,000-task agentic deployment, ∼562 tasks fail silently without external constraint versus ∼214 with constraint.
- Evidence that external metacognitive scaffolding, not better self-knowledge, is the path to safer agentic systems.

Table 1 positions MIRROR relative to prior work.

2 Related Work

LLM Calibration. Kadavath et al. [2022] showed that LLMs can predict their own answer correctness above chance on knowledge-intensive tasks. Kapoor et al. [2024] demonstrated that fine-tuning is needed for reliable calibration. Steyvers et al. [2025] found that metacognitive ability is task-specific, not general—a finding consistent with MIRROR's domain-level analysis. MIRROR differs from all prior calibration work in testing whether calibration *transfers to action-selection*: we measure the gap between knowing and doing, not the accuracy of knowing alone.

LLM Self-Knowledge and Situational Awareness. SAD [Laine et al., 2024] measures whether LLMs know what they are—identity, lifecycle stage, deployment context. Barkan et al. [2025] tested capability prediction on SWE-bench tasks, finding systematic overconfidence. Xiong et al. [2024] found that LLMs are often overconfident when expressing numerical probability estimates. MIRROR is complementary: SAD diagnoses awareness; MIRROR diagnoses the gap between awareness and behavior.

Agentic Evaluation. AgentBench [Liu et al., 2024], OSWorld [Xie et al., 2024], and GAIA [Mialon et al., 2023b] measure task success rate in agentic settings. These benchmarks evaluate execution

quality conditional on the decision to act. Concurrent work on agentic confidence calibration [Anonymous, 2026a] measures calibration during multi-step trajectories using trajectory-level features. MIRROR differs in measuring metacognitive calibration *before* deployment and testing whether pre-deployment assessment predicts agentic outcomes. More broadly, MIRROR inverts the standard framing: we measure the *decision* itself, treating task execution quality as a downstream variable and calibration-informed decision-making as the primary outcome.

Metacognition in AI. Metacognitive prompting [Wang and Zhao, 2024] uses metacognitive strategies to improve LLM performance. Augmented language model surveys [Mialon et al., 2023a] document the tool-use landscape but do not address when tool use *should* occur. Concurrent work on metacognition in large reasoning models [Anonymous, 2026b] tests difficulty awareness, confidence adjustment, and strategy revision across reasoning models. MIRROR extends this by connecting metacognitive assessment to agentic deployment through the Experiment 9 escalation design, providing the first benchmark-scale empirical test linking metacognitive assessment to agentic deployment outcomes.

3 The MIRROR Benchmark

MIRROR decomposes LLM metacognition into four levels of increasing cognitive complexity, inspired by the hierarchical structure of human metacognitive monitoring and control [Flavell, 1979, Nelson and Narens, 1990]. At each level, we measure behavior through five independent channels to prevent format-specific gaming.

3.1 Metacognitive Levels

Level 0—Atomic Self-Knowledge (Calibration). Can the model accurately estimate $P(\text{correct})$ for a given question in a given domain? This is the best-studied level [Kadavath et al., 2022, Kapoor et al., 2024] and serves as MIRROR's foundation. Experiment 1 establishes per-model, per-domain calibration profiles across 8 cognitive domains.

Level 1—Cross-Domain Transfer. Does calibration in domain A influence behavior on tasks that implicitly require domain A skills? A model that knows it is weak at spatial reasoning (Level 0) should show increased caution on a trip-planning task that requires spatial reasoning, even though the surface framing is "travel advice." Experiments 2 and 3 test this transfer.

Level 2—Compositional Prediction. Can the model predict its performance on tasks requiring *combinations* of skills? A model that is 80% accurate on arithmetic and 70% on coding should estimate its accuracy on a task requiring both—but if self-knowledge is atomic rather than compositional, this prediction will fail. Experiment 3 tests this directly.

Level 3—Adaptive Self-Regulation. Does the model translate self-knowledge into appropriate action? This is the critical gap between metacognitive *monitoring* (knowing your limits) and metacognitive *control* (acting on that knowledge)—a distinction well-established in human cognition [Metcalfe and Finn, 2008]. Experiments 4, 5, 6, and 9 test whether self-knowledge drives behavior change, feedback adaptation, ecosystem interactions, and agentic decision-making.

We hypothesize these levels form an empirical difficulty ordering (L0 easiest, L3 hardest). Whether this ordering holds—and where it breaks—is itself a testable prediction.

3.2 Behavioral Measurement Channels

A single confidence elicitation format (*e.g.*, verbal "rate your confidence 1–10") is vulnerable to prompt-specific behavior. MIRROR measures metacognitive behavior through five independent channels, each requiring a separate API call with a fresh context:

1. **Wagering**—the model stakes points proportional to its confidence, with scoring that penalizes overconfident wrong answers.
2. **Opt-out**—the model may decline to answer. Skip rate should correlate with error rate.
3. **Difficulty selection**—given two questions (easy and hard), the model chooses which to attempt.

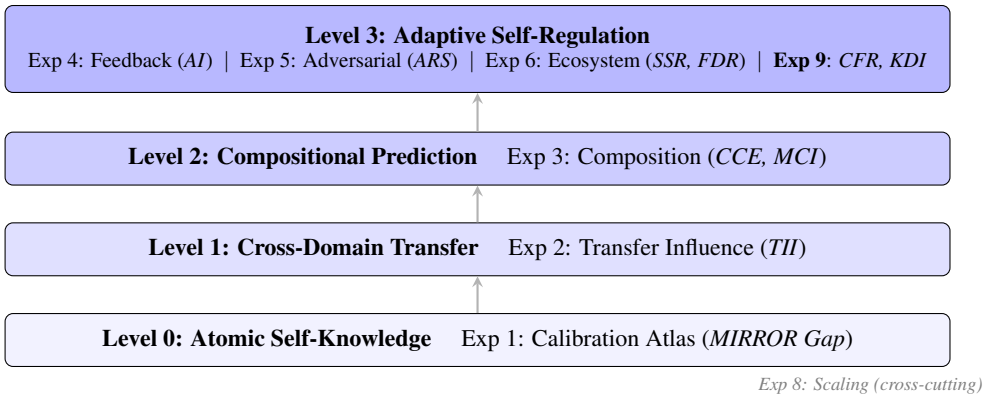

Figure 1: The MIRROR experimental hierarchy. Each level builds on measurements from the level below. Experiment 1 produces per-model calibration profiles consumed by all subsequent experiments. The capstone (Experiment 9) tests whether self-knowledge translates to appropriate agentic decisions.

4. **Tool delegation**—the model may invoke a calculator, search engine, or flag for human review rather than answering directly.
5. **Natural language signals**—hedging language, uncertainty markers, and response length, extracted from unconstrained free-form responses.

The **Metacognitive Convergence Index (MCI)** measures agreement across channels: do all five channels agree about which questions are hard for this model? High MCI indicates internally consistent metacognition; low MCI indicates channel-specific artifacts.

3.3 Experiments

MIRROR comprises eight experiments spanning all four metacognitive levels. Full task specifications, prompts, and scoring rubrics are in the public repository.

Experiment 1: Self-Knowledge Atlas (Level 0). 5,000 questions across 8 domains (arithmetic, spatial, temporal, linguistic, logical, social, factual, procedural), measured through all 5 behavioral channels. 16 models evaluated. Primary metrics: MIRROR gap, MCI.

Experiment 2: Cross-Domain Transfer (Level 1). $\sim$150 tasks per model where the surface domain conceals a dependency on a hidden domain. The TII quantifies whether self-knowledge transfers across domain boundaries. 11 models; TII range: 0.019–0.175 (weak-to-minimal transfer).

Experiment 3: Compositional Self-Prediction (Level 2). $\sim$25 multi-domain tasks per model. The CCE measures predicted vs. actual composite accuracy. 15 models evaluated. Primary finding: MCI = 0.000 across all models.

Experiment 4: Adaptation Crucible (Level 3). 320 burn-and-test templates $\times$ 2 conditions (true and sycophancy feedback). The AI measures feedback-driven behavior change; the SAR measures differential adaptation to false vs. true feedback. 15 models; AI $\approx$ 0 (feedback immunity); SAR range: -1.52 to 2.97.

Experiment 5: Adversarial Robustness (Level 3). 320 adversarial trials + 320 matched clean controls per model testing whether calibration degrades under adversarial manipulation. 14 models; ARS = 0.937–0.991 (robust).

Table 2: Model roster. All models are instruction-tuned variants accessed via API. Parameter counts are approximate where not officially disclosed.

Model	Lab	Params	Arch.	API	Experiments
`deepseek-r1`	DeepSeek	671B	Dense	DeepSeek	1–6, 9
`deepseek-v3`	DeepSeek	671B	Dense	DeepSeek	1–6
`gemini-2.5-pro`	Google	>1T	Dense	Google AI	1, 4, 6
`gemma-3-12b`	Google	12B	Dense	Google AI	1, 3–6
`gemma-3-27b`	Google	27B	Dense	NVIDIA NIM	1–6, 9
`gpt-oss-120b`	OpenAI	120B	Dense	NVIDIA NIM	1–6, 9
`kimi-k2`	Moonshot AI	200B	Dense	NVIDIA NIM	1–6, 9
`llama-3.1-405b`	Meta	405B	Dense	NVIDIA NIM	1–4, 6, 8–9
`llama-3.1-70b`	Meta	70B	Dense	NVIDIA NIM	1–6, 8–9
`llama-3.1-8b`	Meta	8B	Dense	NVIDIA NIM	1–6, 8–9
`llama-3.2-3b`	Meta	3B	Dense	NVIDIA NIM	1, 3–6
`llama-3.3-70b`	Meta	70B	Dense	NVIDIA NIM	1–6, 8–9
`mistral-large`	Mistral	675B	Dense	NVIDIA NIM	1–3, 5–6, 9
`mixtral-8x22b`	Mistral	193B	MoE (8x22B)	NVIDIA NIM	1, 3–6
`phi-4`	Microsoft	14B	Dense	NVIDIA NIM	1–6, 9
`qwen3-next-80b`	Alibaba	80B	Dense	NVIDIA NIM	1, 3–6

Experiment 6: Ecosystem Effect (Level 3). (a) Sycophancy in multi-agent trust scenarios (SSR). (b) Flawed premise detection (FDR: 0.000–1.000). (c) Value robustness under incremental pressure. 16 models evaluated.

Experiment 8: Scaling Analysis (Levels 0–3). Within-family comparison across Llama 3.1 8B $\rightarrow$ 70B $\rightarrow$ 405B and a generation comparison (Llama 3.1-70B vs. 3.3-70B). Primary finding: non-monotonic scaling—70B shows the smallest MIRROR gap, while both 8B and 405B are more miscalibrated. Calibration improves with generation (3.3 $>$ 3.1 at matched size) but not monotonically with parameters.

Experiment 9: The Knowing-Doing Gap (Level 3+). 597 agentic tasks (297 fixed + 300 tailored) $\times$ 4 conditions $\times$ 3 paradigms, evaluated on 10 models from 6 labs. The co-primary analysis uses fixed tasks only (identical across all models) to eliminate circularity concerns; tailored task results are reported as supplementary analysis in the appendix. Each task has two components requiring different domain skills. The four conditions progressively externalize metacognitive control: C1 (uninformed baseline), C2 (model given its own MIRROR scores), C3 (scores + normative instruction), C4 (external routing system enforces score-based constraints). Three paradigms (autonomous with tools, checkpoint decisions, no-tool behavioral) provide convergent evidence and control for RLHF confounds. Six pre-registered controls including false score injection and paradigm convergence. Primary metrics: CFR, KDI, Escalation Curve.

3.4 Evaluation Infrastructure

MIRROR is designed for reproducible, low-cost evaluation of new models. The benchmark requires only API access (no weights, no activations, no fine-tuning). Running the full benchmark on a new model requires approximately 8,000 API calls ($\sim$3 hours at 40 requests/minute). All prompts, scoring rubrics, and analysis code are publicly available. The dataset is hosted on HuggingFace with Croissant metadata.

Table 3: Summary statistics for each MIRROR experiment.

Exp.	Description	Models	Items/Model	Primary Metric	M
Exp1	Atomic Self-Knowledge	16	330/model (8 domains x 5 subcats x 8 Qs)	MIRROR Gap	0
Exp2	Cross-Domain Transfer	11	150/model	TII	0
Exp3	Compositional Self-Prediction	15	25/model (mixed tasks)	CCE	0
Exp4	Adaptive Self-Regulation (Burn-a...	15	320/model x 2 conditions	AI	0
Exp5	Adversarial Robustness of Self-K...	14	320/model (adv) + 320/model (clean)	ARS	0
Exp6	Ecosystem Effect (Trust + Flawed...	16	240/model (6a+6b+6c)	FDR	0
Exp8	Scaling Analysis	4	80/model (gap-fill)	Scaling slope	0
Exp9	The Knowing-Doing Gap	10	597 tasks x 4 conditions x 3 paradigms	CFR (C1)	0

4 Experimental Setup

We evaluate 16 models from 8 labs spanning 3B to 671B parameters (Table 2). All 16 models participate in the core experiments (1, 4, 5, 6); subsets participate in experiments with specific requirements: 11 models in Experiment 2 (requires cross-domain task infrastructure), 15 in Experiment 3, 10 in Experiment 9 (requires complete Experiment 1 profiles for tailored task generation), and 3 in Experiment 8 (Llama family scaling analysis). Two additional models were excluded: `command-r-plus` (API removed mid-collection) and `qwen-3-235b` (100% API failure rate on the NIM endpoint). All experiments use `temperature=0` for reproducibility. Infrastructure: NVIDIA NIM free tier (primary, for Llama, Mistral, Gemma, Phi-4, Kimi-K2), DeepSeek API (deepseek-r1/v3), and Google AI Studio (gemini-2.5-pro, gemma-3-12b). Concurrency: 32 parallel calls per model. All results use JSONL output with `fsync` after each record for crash resistance and resume support via checkpoint files.

Table 3 summarizes the scale and primary metrics for each experiment.

5 Results

5.1 Main Results

Table 4 reports all primary metrics across 16 models and 8 experiments. Three patterns are immediately visible. First, the MIRROR gap is universally positive (mean = 0.170, range 0.074–0.267): all models are overconfident in the wagering channel relative to natural accuracy. Second, CFR in the uninformed condition ranges from 0.387 to 0.907—even the best model fails on 39% of weak-domain agentic tasks when proceeding autonomously. Third, KDI is negative for 9 of 10 models evaluated in Experiment 9, confirming that self-knowledge systematically fails to translate into appropriate action.

5.2 Finding 1: Compositional Self-Prediction Fails Universally

Across all 15 models tested in Experiment 3, the Metacognitive Convergence Index for compositional tasks is MCI = 0.000. Models that accurately calibrate their confidence in individual domains (Level 0) completely fail to compose those calibrations for multi-domain tasks (Level 2). This is not a gradual degradation—it is a categorical failure. A model that knows "I am 80% accurate on arithmetic" and "I am 70% accurate on coding" cannot predict its accuracy on a task requiring both.

The Compositional Calibration Error (CCE) ranges from 0.500 (gemma-3-12b, llama-3.1-405b) to 0.943 (mistral-large), meaning mistral-large's expressed composite confidence sits 0.94 probability units above floor-level actual performance.

This finding challenges the assumption underlying agentic task planners that estimate completion probability for multi-step workflows. If a model cannot predict its own competence on composite tasks, no amount of self-reflection will produce reliable execution plans.

Table 4: Main results across all experiments for 16 models. Nat.Acc = natural accuracy (Exp1), Wag.Acc = wagering accuracy (Exp1), M.Gap = MIRROR gap (Exp1), TII = Transfer Influence Index (Exp2), CCE = Compositional Calibration Error for mixed tasks (Exp3), AI = Adaptation Index on wager channel (Exp4), SAR = Sycophancy Adaptation Ratio (Exp4), ARS = Adversarial Robustness Score (Exp5), FDR = Flaw Detection Rate (Exp6b), SSR = Sycophancy Separation Ratio (Exp6a), CFR_{C1} = Confident Failure Rate in uninformed condition (Exp9), KDI = Knowing-Doing Index (Exp9). "—" indicates the model was not evaluated in that experiment.

Model	Nat.Acc	Wag.Acc	M.Gap	TII	CCE	AI	SAR	ARS	FDR	SSR	CFR_{C1}	KDI
deepseek-r1	0.437	0.569	0.139	0.175	0.511	-0.0652	0.466	0.961	0.133	0.02	0.387	-0.028
deepseek-v3	0.392	0.659	0.267	0.167	0.513	0.0071	0.461	0.983	1.000	0.97	—	—
gemini-2.5-pro	0.235	0.298	0.074	—	—	0.0131	2.969	—	0.000	—	—	—
gemma-3-12b	0.449	0.609	0.160	—	0.500	0.1717	0.890	0.989	1.000	6.17	—	—
gemma-3-27b	0.487	0.626	0.150	0.154	0.667	0.0000	—	0.960	1.000	1.91	0.688	-0.271
gpt-oss-120b	0.280	0.441	0.161	0.173	0.688	-0.0001	-0.000	0.991	1.000	1.09	0.657	-0.136
kimi-k2	0.484	0.675	0.191	0.019	0.887	0.0035	1.125	0.984	0.926	0.97	0.625	-0.255
llama-3.1-405b	0.392	0.464	0.102	0.062	0.500	0.0000	—	—	1.000	3.10	0.613	-0.356
llama-3.1-70b	0.414	0.501	0.212	0.049	0.551	0.0836	2.013	0.948	0.838	2.10	0.550	-0.168
llama-3.1-8b	0.388	0.532	0.186	0.107	0.807	0.0000	—	0.937	1.000	4.42	0.787	-0.139
llama-3.2-3b	0.426	0.577	0.179	—	0.577	-0.0781	2.120	0.982	1.000	3.54	—	—
llama-3.3-70b	0.499	0.767	0.267	0.100	0.797	-0.0016	-1.523	0.974	0.825	2.33	0.547	0.049
mistral-large	0.349	0.462	0.113	0.129	0.943	—	—	0.946	0.600	2.11	0.907	-0.080
mixtral-8x22b	0.460	0.593	0.136	—	0.647	-0.0252	0.363	0.984	1.000	1.60	—	—
phi-4	0.279	0.483	0.222	0.068	0.906	-0.0074	-0.000	0.960	1.000	3.30	0.721	-0.304
qwen3-next-80b	0.486	0.638	0.152	—	0.882	0.0750	0.958	0.956	1.000	1.73	—	—

5.3 Finding 2: The Knowing-Doing Gap

The central finding of MIRROR is a functional dissociation between metacognitive monitoring and metacognitive control. Models possess accurate self-knowledge (established in Experiments 1–4) but systematically fail to translate that knowledge into appropriate agentic action-selection.

The Escalation Curve. We test this through four conditions that progressively externalize metacognitive control ($n = 10$ models, all paradigms; see Figure 2):

Total CFR reduction C1→C4: **62%** (34.8 percentage points). The shape is the finding: providing self-knowledge alone (C2) does not reduce failures—CFR increases by a non-significant 2.1 pp, suggesting that explicit self-knowledge adds no actionable value. Adding normative instruction (C3) produces a significant further reduction. Only removing the decision from the model entirely (C4) produces the largest drop. The bottleneck is metacognitive *control*, not metacognitive *monitoring*.

Universal Negative KDI. The Knowing-Doing Index (KDI = self-knowledge accuracy − appropriate action rate) is negative for 9 of 10 models, ranging from -0.356 (llama-3.1-405b) to $+0.049$ (llama-3.3-70b). Figure 3 shows per-model values. The lone positive outlier (llama-3.3-70b) is driven by its procedural domain, where nat_acc $= 0.059$ but the model proceeds autonomously on 70% of procedural tasks—the knowing-doing gap at its most extreme.

Null Predictive Result (Pre-registered). The MIRROR calibration gap does not independently predict agentic CFR beyond raw accuracy: Pearson $r = -0.100$, partial r (controlling for natural accuracy) $= -0.044$, $p = 0.557$. This null was pre-registered as Contingency C1. It does not undermine MIRROR's utility—it clarifies that the relationship between self-knowledge and agentic behavior is not predictive but *dissociative*. The fix is not better self-knowledge but external metacognitive scaffolding.

Deployment Implications. In a 1,000-task agentic deployment: ∼562 tasks fail silently under C1 (the model proceeds on weak domains despite having internal self-knowledge). Providing the model with its own MIRROR scores (C2) increases this to ∼583—it gets slightly *worse*. Only external constraint (C4) brings failures down to ∼214, a 62% reduction. Every lab deploying agentic LLMs should evaluate metacognitive *control*, not just metacognitive monitoring.

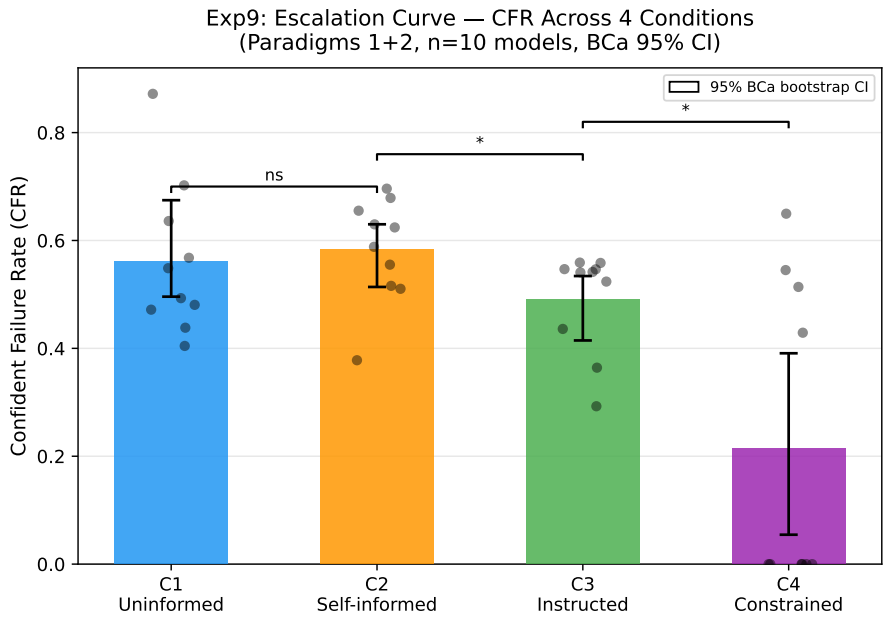

Figure 2: CFR escalation curve across four Experiment 9 conditions (mean $\pm$ 95% BCa bootstrap CI, $n = 10$ models). Self-knowledge alone (C1$\rightarrow$C2) does not reduce CFR; all 34.8 pp of improvement comes from external structure (C2$\rightarrow$C4). External constraint drives the reduction.

5.4 Supporting Findings

Feedback Immunity (Exp 4). AI ≈ 0 across most models—self-knowledge is not updated by experience. The two models that do adapt (deepseek-r1, llama-3.1-70b) show a domain-polarity signature: positive adaptation in factual/temporal domains, negative in procedural/social.

Adversarial Robustness (Exp 5). ARS = 0.937–0.991 across all 14 models tested. Metacognitive calibration is robust to framing attacks—models do not become more overconfident under adversarial pressure.

Flaw Detection (Exp 6b). Extreme inter-model variation: FDR ranges from 0.000 (gemini-2.5-pro—never flags flaws) to 1.000 (gemma-3-12b). deepseek-r1 achieves the lowest CFR in Experiment 9 (0.387, most cautious) but has FDR = 0.133 (worst flawed-premise detector among non-outlier models)—cautious in action but poor at epistemic judgment. Models from the same lab can differ dramatically.

Non-Monotonic Scaling (Exp 8). Within the Llama 3.1 family, 70B shows the smallest MIRROR gap; both 8B and 405B are more miscalibrated. The generation comparison (3.1-70B vs. 3.3-70B) shows capability improvement (+20.2 pp natural accuracy) without closing the calibration gap (+0.010 gap increase). Metacognitive calibration does not scale monotonically with parameters.

Weak Transfer (Exp 2). TII = 0.019–0.175 across 11 models. Self-knowledge is domain-atomic—it does not meaningfully cross domain boundaries. Verbal transfer scores (models correctly identifying their weakest domain in cross-domain tasks) are consistently low (0.039–0.140).

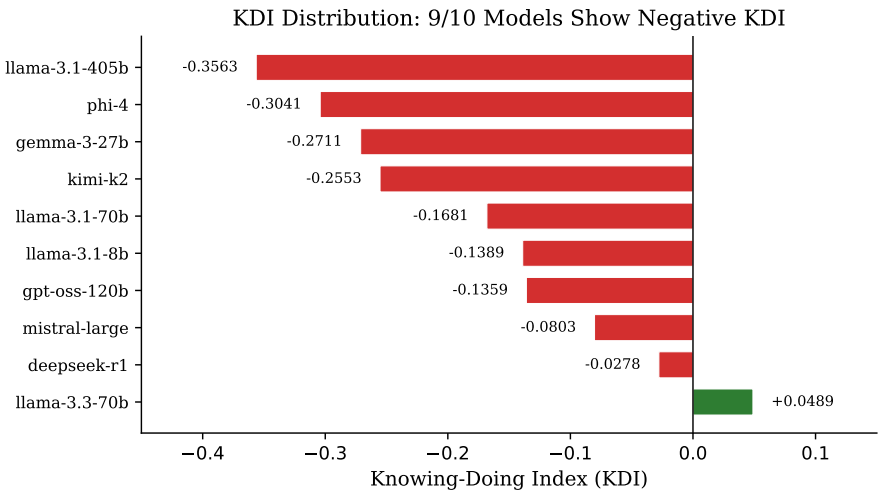

Figure 3: KDI distribution across 10 models. Nine models show uniformly negative KDI, indicating systematic failure to translate self-knowledge into appropriate action. The lone positive outlier (llama-3.3-70b, $KDI = +0.049$) reflects extreme procedural-domain miscalibration.

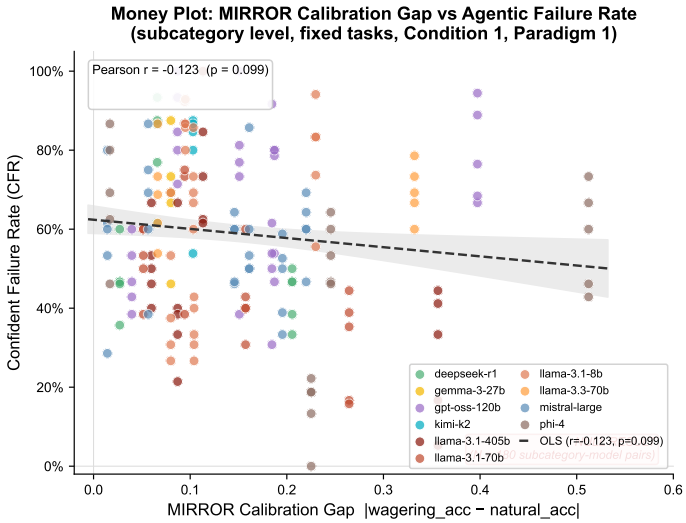

Figure 4: MIRROR gap vs. CFR at the subcategory level. The shown P1 fixed-task correlation ($r = -0.123$, $p = 0.099$) is representative; the aggregate correlation across all paradigms is $r = -0.100$. Calibration gap does not independently predict agentic failure. Raw accuracy ($r = -0.386$, $p < 0.001$) is the dominant predictor.

6 Discussion and Limitations

6.1 Implications for Agentic System Design

The knowing-doing gap reframes the field's approach to agentic safety. Current practice focuses on improving calibration—making models more accurate in their self-assessment. MIRROR shows this is necessary but insufficient: even models with accurate domain-specific self-knowledge fail to translate that knowledge into appropriate action-selection in 56.2% of weak-domain cases (C1 baseline).

The fix is architectural, not epistemic. External routing that uses MIRROR domain scores as decision inputs reduces CFR by 62%. We recommend that every lab deploying agentic systems evaluate metacognitive *control* (not just monitoring) before deployment. MIRROR-informed routing

offers a practical intervention: use per-domain calibration scores to determine which task components require external constraints, tool delegation, or human review.

Additional analyses of the sycophancy-feedback asymmetry (Appendix C) and RLHF confound analysis (Appendix D) provide further context for these findings.

6.2 Limitations

API-only evaluation. MIRROR requires only API access, enabling scale but preventing mechanistic analysis. Whether the knowing-doing gap reflects an information deficit or a control failure at the representation level remains open for hidden-state probing.

No human annotations. MIRROR is fully automated: questions generated and verified by LLM ensembles. This enables scale (250K instances) but prevents human–LLM comparison.

Ecological validity. Experiment 9 uses constructed agentic scenarios, not production deployments.

Model coverage. 16 models from 8 labs. Findings may not generalize to proprietary models (GPT-4o, Claude 3.5) not tested. Experiment 2 is limited to 11 models; Experiment 6c is partial for 3 models due to API unavailability.

Null predictive result. MIRROR gap does not predict CFR independently of accuracy (partial $r = -0.044$). The contribution is diagnostic (revealing the knowing-doing gap), not predictive. The 270 planned tailored tasks not completed before analysis would have increased statistical power.

MCI design artifact. The pre-registered MCI requires strong-strong and weak-weak composite trials. Task generation rarely produced both simultaneously, rendering MCI = 0 partially a design artifact. CCE is the recommended primary Level 2 metric.

6.3 Future Work

Future work includes mechanistic probing of the knowing-doing gap via hidden-state analysis on open-weight models, human baseline comparisons, behavioral fine-tuning on MIRROR data, multi-agent cascade analysis, and production deployment of MIRROR-informed routing.

7 Conclusion

We have introduced MIRROR, a hierarchical benchmark for metacognitive calibration in LLMs, and reported results across eight experiments spanning 16 models from 8 labs. The central finding is precise: models possess domain-specific self-knowledge that accurately identifies weak domains, but this self-knowledge does not transfer across domains, does not compose correctly, does not update from feedback, and does not translate into appropriate agentic action. The Knowing-Doing Index is negative for 9 of 10 models. Scale does not resolve this.

What works is external structure: architectural constraint reduces the Confident Failure Rate by 62%, from 0.562 to 0.214. The implication for AI safety is that trust in an LLM's self-reported uncertainty—a common design assumption in agentic systems—is insufficient. Calibration must be measured externally, then enforced architecturally. MIRROR provides the measurement instrument.

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A Scaling Analysis

Figure 5 shows the Experiment 8 scaling analysis. Within the Llama 3.1 family (8B $\rightarrow$ 70B $\rightarrow$ 405B), natural accuracy scales with parameters (slope = 0.028/log-unit, $R^2 = 0.823$) but the MIRROR gap does not (slope ≈ 0, $R^2 = 0.000$). The generation comparison (3.1-70B vs. 3.3-70B) shows capability improvement (+20.2 pp natural accuracy, +21.2 pp wagering accuracy) without closing the calibration gap (+0.010 MIRROR gap increase).

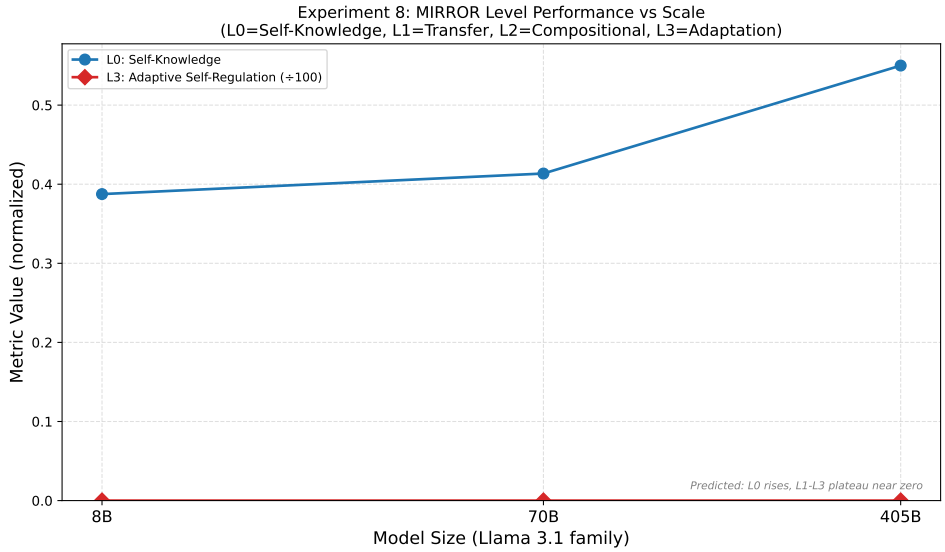

Figure 5: Scaling analysis: MIRROR metrics across model size (Llama 3.1 family) and generation (3.1 vs. 3.3 at 70B). Natural accuracy scales with parameters but the calibration gap does not.

B Per-Paradigm Escalation

Figure 6 shows the CFR escalation curve broken down by paradigm. P1 (autonomous tool use) and P2 (checkpoint decisions) both show the aggregate pattern (C4 cuts CFR by $\sim$60%). P3 (behavioral, no tools) has no C4 condition. The aggregate 62% reduction is driven by both tool-enabled paradigms.

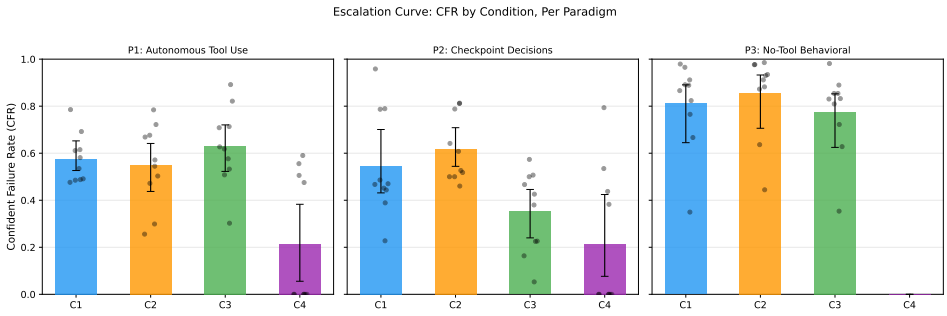

Figure 6: Per-paradigm CFR escalation curves. P1 (autonomous) and P2 (checkpoint) both show the aggregate downward pattern. P3 (no tools) has no C4 condition.

C Sycophancy-Feedback Asymmetry

Experiments 4 and 6 together establish a striking asymmetry. Models do not update their calibration channels from accurate performance feedback (AI ≈ 0, Exp. 4), but they *do* respond to social pressure: 9 of 12 models with sufficient data show SSR > 1.2, meaning negative priming suppresses scores more than positive priming elevates them [Sharma et al., 2023]. The asymmetry is this: models are responsive to evaluator affect and unresponsive to data. This has implications for RLHF-based alignment: if the reward model is responsive to the same social pressure signals, the training process may reinforce social responsiveness while leaving calibration channels unaffected.

D Paradigm Convergence and RLHF Confounds

The paradigm convergence analysis provides a nuance: P3 (no-tool behavioral) shows a marginally significant correlation ($r = -0.332$, $p = 0.039$) between MIRROR gap and behavioral signals, while P1 ($r = -0.114$, $p = 0.489$) and P2 ($r = -0.277$, $p = 0.088$) do not. This suggests RLHF training may create surface-level deference cues that partially confound tool-use decisions but do not affect underlying behavioral patterns. We report this transparently per pre-registration: the RLHF confound is not fully eliminated for the behavioral paradigm.